

Assignment 4 – FIR DAC

- A. Design a **2nd-order feed-forward continuous-time** sigma delta modulator for 20kHz BW with a **1-bit** quantizer and Ideal NRZ DAC, achieving peak **SQNR of 100dB**.
- Putting a 3-tap FIR filter **after the modulator** (outside the loop), show the FFT of the FIR filter output, analyze the spectrum in the comments. [2 points]
 - Moving the FIR filter after the QNT (into the loop), show the FFT of the modulator output, explain the result in the comments. [1 point]
 - Break the loop, show the **impulse responses** of the signal chain : **DAC=>LF** and **DAC=>FIR=>LF**. [2 points]
 - **Compensate the loop** based on the result of impulse responses, show the modulator output FFT again. [3 points]
 - Explain the difference between modulators w/i and w/o FIR DAC (MSA, STF). [2 points]
- **Submit your answers before the class on Tuesday Jun 2nd, 10:30.**
 - **Instructions about submitting assignments are on Brightspace.**